

# Algebra 1

Part 1





## About the Course

Students discover and practice algebraic concepts through real-world narratives and problem-solving.

This topic is included in the following course(s): Mathematics: Grade 8

### Mathematics: Grade 8

For students in elementary school, we recommend RightStart Math (Second Edition), because a) it adheres to Mason's principle that the idea always comes before the algorithm, b) it reflects the very best current research in how children learn mathematics, and c) it offers much-needed support for teachers whose own math instruction may have been lacking. Students in middle school take RightStart Geometry and also add in Pre-Algebra (Hands-On Equations) and Jacobs' Elementary Algebra.

The lessons should be done in the morning and games in the afternoon. Both mental math and games are critical parts of math instruction, so do not skip them. These lessons are paced out to help keep you on track. There are a few catch-up days built in and can be used to finish incomplete work or to review concepts. The general goal should be to complete one lesson per day. It is important that you complete one level per year, even if that means continuing during breaks, as moving slower can affect high school credits later.



## Placement & Combining Tips

Even students in Grade 8 who have not taken Pre-Algebra should take this course to keep on pace for earning high school credits.

Prerequisite: RightStart F or mastery of basic arithmetic, fractions, decimals, and percents.



## Scheduling

| GRADE | SCHEDULE INFO.                           | BOOKS                                                                                 |
|-------|------------------------------------------|---------------------------------------------------------------------------------------|
| 8     | Algebra 1: Part 1<br>3 times/week 40 min | Elementary Algebra Student Book<br>Elementary Algebra Teacher Guide (Revised Edition) |

### Sample Weekly View

| Day 1                                       | Day 2                                       | Day 3                                       | Day 4             | Day 5             |
|---------------------------------------------|---------------------------------------------|---------------------------------------------|-------------------|-------------------|
| <b>Mathematics: Grade 8</b>                 |                                             |                                             |                   |                   |
| Algebra 1: Part 1<br>Math Games: Grades 1-8 | Geometry: Grade 8<br>Math Games: Grades 1-8 | Algebra 1: Part 1<br>Math Games: Grades 1-8 | Geometry: Grade 8 | Algebra 1: Part 1 |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the

time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Mathematics: Grade 8

☐ If you downloaded the RightStart ebook, print out enough copies of each page and put them in a binder for easy access.

☐ Organize math manipulatives for easy access.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Algebra 1: Part 1



Elementary Algebra Student Book



Elementary Algebra Teacher Guide (Revised Edition)



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

Related Course Quick Links

∞ [Foundations \(See Section 10: Math\)](#)

Click THIS text  
or scan the QR  
code for links.



# Algebra 1: Part 1

## How To Teach



### Read

The RightStart Math Level G materials are written to the student and may be done independently. However, you may still find it very helpful to work through the lessons to familiarize yourself with the concepts.



### Prepare

For Pre-Algebra, it is highly recommended that students be guided by a teacher during these lessons. Teachers may watch the video lessons to learn the concepts and how to teach them to students. If you choose to have your student work independently using the video lessons, you may still find it very helpful to work through the lessons yourself to familiarize yourself with the concepts. Teachers must stay apprised of how students are progressing and be ready to step in and help, since problems can compound quickly. If you are working through the lessons ahead of your student, note the places where you have to stop and think, because those are probably going to be places where students are going to be challenged, as well.



### Growth Mindset

Resist the temptation to fall back on just teaching a memorizable procedure when the student struggles with a concept. Growth happens in the struggle. Work on resilience by modeling a growth mindset.



### Leave & Return

When your student hits a wall during a lesson, stop and move on to the next subject in your day or take a break. Explain that even when you stop working on a problem, your brain continues to work on it while you do other things, and you don't even know it. Many times, when you come back to the problem the next day, it comes more easily.



### Connect

Look for ways to bring math into everyday life and conversation "by the way" - through cooking, gardening, house projects, health, games, etc. Students should discover how math is useful and beautiful, and how it helps us explain and navigate the world.

# Algebra 1: Part 1

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## Term 1

### WEEK 1 40m Algebra 1: Part 1 - Lesson 1

*A Number Trick*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Introduction p.1-3

"Think of a " - "such as these."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Exercises p.3-4

### WEEK 1 40m Algebra 1: Part 1 - Lesson 2

*Addition*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 1 p.6-8

"Soon after" - "any further."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.8-9

### WEEK 1 40m Algebra 1: Part 1 - Lesson 3

*Subtraction*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 2 p.11-12

"The month of" - "from the first."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.12-14

### WEEK 2 40m Algebra 1: Part 1 - Lesson 4

*Multiplication*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 3 p.15-16

"Learning the" - "applied to division."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.17-18

# Algebra 1: Part 1

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## Term 1

### WEEK 2 □ 40m Algebra 1: Part 1 - Lesson 5

*Division*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 4 p.20-21

"If two people" - "to give x."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.21-22

### WEEK 2 □ 40m Algebra 1: Part 1 - Lesson 6

*Raising to a Power*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 5 p.24-25

"It is an" - "product of n x's."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.25-27

### WEEK 3 □ 40m Algebra 1: Part 1 - Lesson 7

*Zero and One*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 6 p.28-29

"Although the Alexandrian" - " $x/1 = x$ ."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.30-31

### WEEK 3 □ 40m Algebra 1: Part 1 - Lesson 8

*Several Operations*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 7 p.32-34

"What do you think" - " $4 - 4 = 0$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.35-36

# Algebra 1: Part 1

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## Term 1

### WEEK 3 40m Algebra 1: Part 1 - Lesson 9

*Parentheses*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 8 p.38-40

"Parentheses are" - " $10 + 3 = 13$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.40-42

### WEEK 4 40m Algebra 1: Part 1 - Lesson 10

*The Distributive Rule*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations

Lesson 9 p.43-44

"The oldest game" - " $x^2 - x$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.45-46

### WEEK 4 40m Algebra 1: Part 1 - Lesson 11

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 4 40m Algebra 1: Part 1 - Lesson 12

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Ch.1: Fundamental Operations

Set 1 (every other question) p.50-51

### WEEK 5 40m Algebra 1: Part 1 - Lesson 13

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Ch.1: Fundamental Operations

Set 2 (every other question) p.51-52

# Algebra 1: Part 1

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## Term 1

### WEEK 5 40m Algebra 1: Part 1 - Lesson 14

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 5 40m Algebra 1: Part 1 - Lesson 15

*An Introduction to Functions*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.2: Functions and Graphs

Lesson 1 p.54-55

"How does a person" - "age into account."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.55-57

### WEEK 6 40m Algebra 1: Part 1 - Lesson 16

*The Coordinate Graph*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.2: Functions and Graphs

Lesson 2 p.59-60

"According to" - "a simple pattern."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.60-62

### WEEK 6 40m Algebra 1: Part 1 - Lesson 17

*More on Functions*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.2: Functions and Graphs

Lesson 3 p.63-64

"Roger Bannister" - "in the curve."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.65-66

### WEEK 6 40m Algebra 1: Part 1 - Lesson 18

*Direct Variation*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.2: Functions and Graphs

Lesson 4 p.68-69



# Algebra 1: Part 1

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## Term 1

"At any given" - "intersects the origin."

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.70-71

### WEEK 7 40m Algebra 1: Part 1 - Lesson 19

*Linear Functions*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.2: Functions and Graphs

Lesson 5 p.73-74

"The deepest mine" - "a straight line."

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.75-76

### WEEK 7 40m Algebra 1: Part 1 - Lesson 20

*Inverse Variation*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.2: Functions and Graphs

Lesson 6 p.78-79

"The time that" - " $y = 120/x$ "

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.80-81

### WEEK 7 40m Algebra 1: Part 1 - Lesson 21

*Catch-Up Day*

Materials: Jacobs Algebra

### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore

Extra Helpings.

### WEEK 8 40m Algebra 1: Part 1 - Lesson 22

*Summary and Review*

Materials: Jacobs Algebra

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Ch.2: Functions and Graphs

Set 1 (every other question) p.84-86

# Algebra 1: Part 1

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## Term 1

### WEEK 8 40m Algebra 1: Part 1 - Lesson 23

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.2: Functions and Graphs  
Set 2 (every other question) p.86-88

### WEEK 8 40m Algebra 1: Part 1 - Lesson 24

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 9 40m Algebra 1: Part 1 - Lesson 25

*The Integers*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 1 p.90-91  
"The substance" - "in several lessons."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.91-92

### WEEK 9 40m Algebra 1: Part 1 - Lesson 26

*More on the Coordinate Graph*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 2 p.94-95  
"To help people" - "x-axis it is."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.96-97

### WEEK 9 40m Algebra 1: Part 1 - Lesson 27

*Addition*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 3 p.99-100  
"In 1054" - "= -8"

#### → PRACTICE

# Algebra 1: Part 1

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## Term 1

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.101

### WEEK 10 40m Algebra 1: Part 1 - Lesson 28

#### Subtraction

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 4 p.103-104  
"In arithmetic," - "taking away."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.105-106

### WEEK 10 40m Algebra 1: Part 1 - Lesson 29

#### Multiplication

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 5 p.107-109  
"This photograph," - "is always positive."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.109-110

### WEEK 10 40m Algebra 1: Part 1 - Lesson 30

#### Division

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 6 p.111-112  
"It is always possible" - "is always positive."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.112-113

### WEEK 11 40m Algebra 1: Part 1 - Lesson 31

#### Several Operations

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.3: The Integers  
Lesson 7 p.114-115  
"The first official" - "in this lesson."

#### → PRACTICE

# Algebra 1: Part 1

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## Term 1

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.115-117

### WEEK 11 40m Algebra 1: Part 1 - Lesson 32

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 11 40m Algebra 1: Part 1 - Lesson 33

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.3: The Integers  
Set 1 (every other question) p.119-120

### WEEK 12 40m Algebra 1: Part 1 - Lesson 34

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

### WEEK 12 40m Algebra 1: Part 1 - Lesson 35

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

### WEEK 12 40m Algebra 1: Part 1 - Lesson 36

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

# Algebra 1: Part 1

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## Term 2

### WEEK 1 □ 40m Algebra 1: Part 1 - Lesson 37

*The Rational Numbers*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.4: The Rational Numbers

Lesson 1 p.124-126

"This sign" - " $-1/4 > -4$ ."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.126-127

### WEEK 1 □ 40m Algebra 1: Part 1 - Lesson 38

*Absolute Value and Addition*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.4: The Rational Numbers

Lesson 2 p.128-130

"A test given" - " $-6.1$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.131-132

### WEEK 1 □ 40m Algebra 1: Part 1 - Lesson 39

*More on Operations with Rational Numbers*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.4: The Rational Numbers

Lesson 3 p.133-134

"Benjamin Franklin" - " $-3.3$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.135-136

### WEEK 2 □ 40m Algebra 1: Part 1 - Lesson 40

*Approximations*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.4: The Rational Numbers

Lesson 4 p.137-139

"Ask someone" - "that is approximate."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.139-140

# Algebra 1: Part 1

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## Term 2

### WEEK 2 40m Algebra 1: Part 1 - Lesson 41

*More on Graphing Functions*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.4: The Rational Numbers

Lesson 5 p.141-143

"Beaches vary" - "the result shown."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.143-144

### WEEK 2 40m Algebra 1: Part 1 - Lesson 42

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 3 40m Algebra 1: Part 1 - Lesson 43

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Ch.4: The Rational Numbers

Set 1 (every other question) p.147-148

### WEEK 3 40m Algebra 1: Part 1 - Lesson 44

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Ch.4: The Rational Numbers

Set 2 (every other question) p.148-149

### WEEK 3 40m Algebra 1: Part 1 - Lesson 45

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 4 40m Algebra 1: Part 1 - Lesson 46

*Equations*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 1 p.152-153

"The first algebra" - "containing one variable."

# Algebra 1: Part 1

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## Term 2

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.153-154

### WEEK 4 40m Algebra 1: Part 1- Lesson 47

*Inverse Operations*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 2 p.156-158

"One of the" - "result by 5."

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.158-160

### WEEK 4 40m Algebra 1: Part 1- Lesson 48

*Equivalent Equations*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 3 p.162-165

"If the two" - " $19 = 19$ "

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.165-167

### WEEK 5 40m Algebra 1: Part 1- Lesson 49

*Equivalent Expressions*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 4 p.168-171

"One type of" - " $= 4x$ "

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.171-172

### WEEK 5 40m Algebra 1: Part 1- Lesson 50

*More on Solving Equations*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 5 p.173-176

"All the equations" - "So 10 is correct."

### → PRACTICE

# Algebra 1: Part 1

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## Term 2

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.176-177

### WEEK 5 40m Algebra 1: Part 1 - Lesson 51

*Length and Area*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 6 p.179-181

"A baseball diamond" - " $10 + 38 = 48$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)

Set 2 (every other question) p.181-183

### WEEK 6 40m Algebra 1: Part 1 - Lesson 52

*Distance, Rate, and Time*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 7 p.185-186

"It isn't possible" - "miles per hour."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.186-187

### WEEK 6 40m Algebra 1: Part 1 - Lesson 53

*Rate Problems*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.5: Equations in One Variable

Lesson 8 p.189-191

"Several years ago" - " $105 = 105$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)

Set 2 (every other question) p.192-193

### WEEK 6 40m Algebra 1: Part 1 - Lesson 54

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.



# Algebra 1: Part 1

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## Term 2

### WEEK 7 40m Algebra 1: Part 1 - Lesson 55

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.5: Equations in One Variable  
Set 1 (every other question) p.196-198

### WEEK 7 40m Algebra 1: Part 1 - Lesson 56

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.5: Equations in One Variable  
Set 2 (every other question) p.198-200

### WEEK 7 40m Algebra 1: Part 1 - Lesson 57

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 8 40m Algebra 1: Part 1 - Lesson 58

*Equations in Two Variables*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.6: Equations in Two Variables  
Lesson 1 p.202-203  
"So far," - "ordered pairs."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.203-205

### WEEK 8 40m Algebra 1: Part 1 - Lesson 59

*Formulas*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.6: Equations in Two Variables  
Lesson 2 p.206-208  
"Once famous" - " $x = y - 1 / 2$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.208-210

# Algebra 1: Part 1

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## Term 2

### WEEK 8 40m Algebra 1: Part 1 - Lesson 60

*Graphing Linear Equations*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.6: Equations in Two Variables

Lesson 3 p.211-213

"When two dice" - "at the right."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.214-215

### WEEK 9 40m Algebra 1: Part 1 - Lesson 61

*Intercepts*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.6: Equations in Two Variables

Lesson 4 p.216-219

"This is a" - "on neither axis."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.219-221

### WEEK 9 40m Algebra 1: Part 1 - Lesson 62

*Slope*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.6: Equations in Two Variables

Lesson 5 p.222-224

"Donkeys are" - " $-4/2 = -2$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.225-226

### WEEK 9 40m Algebra 1: Part 1 - Lesson 63

*The Slope-Intercept Form*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.6: Equations in Two Variables

Lesson 6 p.227-229

"The amaryllis" - "at the right."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.229-231

# Algebra 1: Part 1

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## Term 2

### WEEK 10 40m Algebra 1: Part 1 - Lesson 64

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 10 40m Algebra 1: Part 1 - Lesson 65

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.6: Equations in Two Variables  
Set 1 (every other question) p.233-234

### WEEK 10 40m Algebra 1: Part 1 - Lesson 66

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.6: Equations in Two Variables  
Set 1 (questions not answered previously) p.233-234

### WEEK 11 40m Algebra 1: Part 1 - Lesson 67

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.6: Equations in Two Variables  
Set 2 (every other question) p.234-235

### WEEK 11 40m Algebra 1: Part 1 - Lesson 68

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 11 40m Algebra 1: Part 1 - Lesson 69

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

# Algebra 1: Part 1

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## Term 2

### WEEK 12 40m Algebra 1: Part 1 - Lesson 70

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

### WEEK 12 40m Algebra 1: Part 1 - Lesson 71

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

### WEEK 12 40m Algebra 1: Part 1 - Lesson 72

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Elementary Algebra

# Algebra 1: Part 1

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## Term 3

### WEEK 1 □ 40m Algebra 1: Part 1 - Lesson 73

*Simultaneous Equations*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 1 p.238-241

"At one time," - "is correct."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.241-242

### WEEK 1 □ 40m Algebra 1: Part 1 - Lesson 74

*Solving by Subtraction*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 2 p.244-245

"Can you solve" - "is correct."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.246-247

### WEEK 1 □ 40m Algebra 1: Part 1 - Lesson 75

*More on Solving by Addition and Subtraction*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 3 p.249-252

"At the time" - "is correct."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.253-254

### WEEK 2 □ 40m Algebra 1: Part 1 - Lesson 76

*Graphing Simultaneous Equations*

□ Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 4 p.255-258

"Suppose that" - " $3 = 3$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.258-260

# Algebra 1: Part 1

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## Term 3

### WEEK 2 □ 40m Algebra 1: Part 1 - Lesson 77

*Inconsistent & Equivalent Equations*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 5 p.261-263

" $x + y = 5$ " - "are equivalent."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.263-266

### WEEK 2 □ 40m Algebra 1: Part 1 - Lesson 78

*Solving by Substitution*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 6 p.267-271

"For two people" - " $11 = 11$ "

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.271-273

### WEEK 3 □ 40m Algebra 1: Part 1 - Lesson 79

*Mixture Problems*

Materials: Jacobs Algebra

#### → READ, NARRATE, & DISCUSS

Ch.7: Simultaneous Equations

Lesson 7 p.274-275

"The greatest" - "a similar way."

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.276-277

### WEEK 3 □ 40m Algebra 1: Part 1 - Lesson 80

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore

Extra Helpings.

### WEEK 3 □ 40m Algebra 1: Part 1 - Lesson 81

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

# Algebra 1: Part 1

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## Term 3

Ch.7: Simultaneous Equations  
Set 1 (every other question) p.280-281

**WEEK 4** **40m Algebra 1: Part 1 - Lesson 82**  
*Summary and Review (cont.)*

Materials: Jacobs Algebra

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.7: Simultaneous Equations  
Set 2 (every other question) p.281-282

**WEEK 4** **40m Algebra 1: Part 1 - Lesson 83**  
*Catch-Up Day*

Materials: Jacobs Algebra

### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

**WEEK 4** **40m Algebra 1: Part 1 - Lesson 84**  
*Large Numbers*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents  
Lesson 1 p.284-286  
"If every chicken" - "Scientific notation"

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.286-287

**WEEK 5** **40m Algebra 1: Part 1 - Lesson 85**  
*A Fundamental Property of Exponents*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents  
Lesson 2 p.289-292  
"The most distant" - " $2.8 \times 10$  to the 12th"

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.292-293

**WEEK 5** **40m Algebra 1: Part 1 - Lesson 86**  
*Two More Properties of Exponents*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents  
Lesson 3 p.294-296  
"Babies are" - "multiply the exponents."

# Algebra 1: Part 1

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## Term 3

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.297-298

### WEEK 5 40m Algebra 1: Part 1 - Lesson 87

*Zero and Negative Exponents*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents

Lesson 4 p.299-302

"When a piano" - " $x$  to the 20th"

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.302-303

### WEEK 6 40m Algebra 1: Part 1 - Lesson 88

*Small Numbers*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents

Lesson 5 p.304-305

"This remarkable" - " $2.7 \times 10$  to the -5th"

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.306-307

### WEEK 6 40m Algebra 1: Part 1 - Lesson 89

*Powers of Products and Quotients*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents

Lesson 6 p.308-310

"One of the" - " $6$  to the 10th"

### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.

Set 1 (odd only)

Set 2 (every other question) p.310-311

### WEEK 6 40m Algebra 1: Part 1 - Lesson 90

*Exponential Functions*

Materials: Jacobs Algebra

### → READ, NARRATE, & DISCUSS

Ch.8: Exponents

Lesson 7 p.313-315

"Suppose that" - "many other topics."

### → PRACTICE



# Algebra 1: Part 1

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## Term 3

Complete Exercises and check solutions. Review any concepts necessary.  
Set 1 (odd only)  
Set 2 (every other question) p.315-316

### WEEK 7 40m Algebra 1: Part 1 - Lesson 91

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 7 40m Algebra 1: Part 1 - Lesson 92

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 7 40m Algebra 1: Part 1 - Lesson 93

*Summary and Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.8: Exponents  
Set 1 (every other question) p.319-320

### WEEK 8 40m Algebra 1: Part 1 - Lesson 94

*Summary and Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.  
Ch.8: Exponents  
Set 2 (every other question) p.321-322

### WEEK 8 40m Algebra 1: Part 1 - Lesson 95

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 8 40m Algebra 1: Part 1 - Lesson 96

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

# Algebra 1: Part 1

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## Term 3

### WEEK 9 40m Algebra 1: Part 1 - Lesson 97

*Midterm Review*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Set 1 (#1-20) and check solutions. Review concepts as necessary.  
p.324

### WEEK 9 40m Algebra 1: Part 1 - Lesson 98

*Midterm Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Set 1 (#21-40) and check solutions. Review concepts as necessary.  
p.324-325

### WEEK 9 40m Algebra 1: Part 1 - Lesson 99

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 10 40m Algebra 1: Part 1 - Lesson 100

*Midterm Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Set 2 (#1-20) and check solutions. Review concepts as necessary.  
p.325-326

### WEEK 10 40m Algebra 1: Part 1 - Lesson 101

*Midterm Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Set 2 (#21-40) and check solutions. Review concepts as necessary.  
p.326

### WEEK 10 40m Algebra 1: Part 1 - Lesson 102

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

# Algebra 1: Part 1

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## Term 3

### WEEK 11 40m Algebra 1: Part 1 - Lesson 103

*Midterm Review (cont.)*

Materials: Jacobs Algebra

#### → PRACTICE

Complete Set 3 (even only) and check solutions. Review concepts as necessary.  
p.327-328

### WEEK 11 40m Algebra 1: Part 1 - Lesson 104

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 11 40m Algebra 1: Part 1 - Lesson 105

*Catch-Up Day*

Materials: Jacobs Algebra

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 12 40m Algebra 1: Part 1 - Lesson 106

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

### WEEK 12 40m Algebra 1: Part 1 - Lesson 107

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

### WEEK 12 40m Algebra 1: Part 1 - Lesson 108

*Exam Week*

Materials: Jacobs Algebra

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Jacobs Elementary Algebra

# Algebra 1: Part 1

## Examination

- |               |                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Term 1</b> | · Jacobs Elementary Algebra: Complete Ch.3 Summary and Review Set 2 (p.121-122, every other question).               |
| <b>Term 2</b> | · Jacobs Elementary Algebra: Complete Ch.6 Summary and Review Set 2 (p.234-235, questions not answered during term). |
| <b>Term 3</b> | · Jacobs Elementary Algebra: Complete Midterm Review Set 3 (p.327-328, odd questions only).                          |